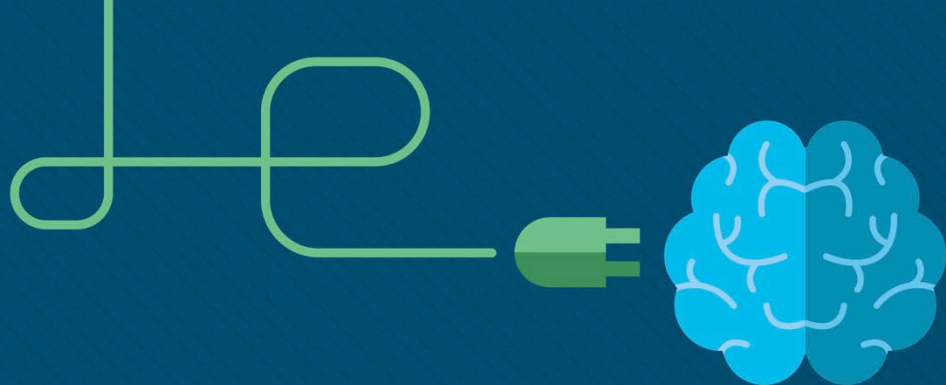




Module 15: IP Static Routing

Switching, Routing, and Wireless Essentials v7.0
(SRWE)



Module Objectives

Module Title: IP Static Routing

Module Objective: Configure IPv4 and IPv6 static routes.

Topic Title	Topic Objective
Static Routes	Describe the command syntax for static routes.
Configure IP Static Routes	Configure IPv4 and IPv6 static routes.
Configure IP Default Static Routes	Configure IPv4 and IPv6 default static routes.
Configure Floating Static Routes	Configure a floating static route to provide a backup connection.
Configure Static Host Routes	Configure IPv4 and IPv6 static host routes that direct traffic to a specific host.

15.1 Static Routes

Types of Static Routes

Static routes are commonly implemented on a network. This is true even when there is a dynamic routing protocol configured.

Static routes can be configured for IPv4 and IPv6. Both protocols support the following types of static routes:

- Standard static route
- Default static route
- Floating static route
- Summary static route

Static routes are configured using the **ip route** and **ipv6 route** global configuration commands.

Static Routes

Next-Hop Options

When configuring a static route, the next hop can be identified by an IP address, exit interface, or both. How the destination is specified creates one of the three following types of static route:

- **Next-hop route** - Only the next-hop IP address is specified
- **Directly connected static route** - Only the router exit interface is specified
- **Fully specified static route** - The next-hop IP address and exit interface are specified

IPv4 Static Route Command

IPv4 static routes are configured using the following global configuration command:

```
Router(config)# ip route network-address subnet-mask { ip-address  
| exit-intf [ip-address] } [distance]
```

Note: Either the *ip-address*, *exit-intf*, or the *ip-address* and *exit-intf* parameters must be configured.

IPv6 Static Route Command

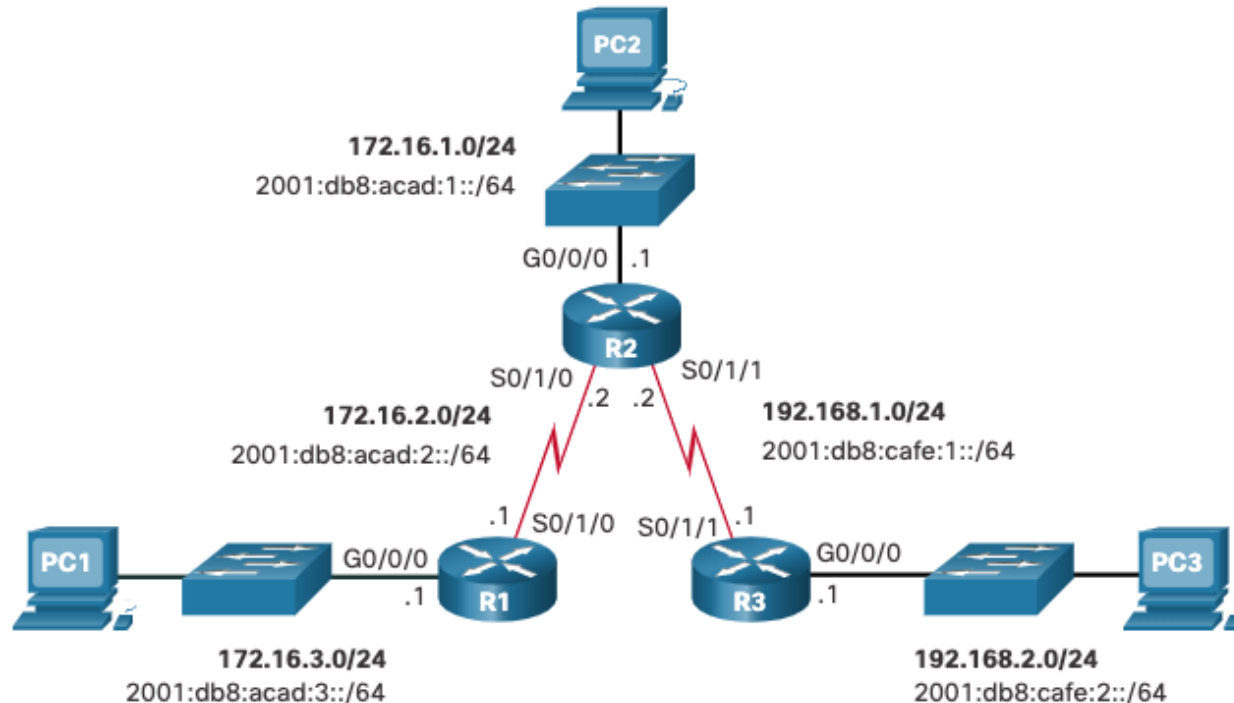
IPv6 static routes are configured using the following global configuration command:

```
Router(config)# ipv6 route ipv6-prefix/prefix-length {ipv6-address  
| exit-intf [ipv6-address]} [distance]
```

Most of parameters are identical to the IPv4 version of the command.

Static Routes Dual-Stack Topology

The figure shows a dual-stack network topology. Currently, no static routes are configured for either IPv4 or IPv6.



Static Routes

IPv4 Starting Routing Tables

- Each router has entries only for directly connected networks and associated local addresses.
- R1 can ping R2, but cannot ping the R3 LAN

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.1/32 is directly connected, Serial0/1/0
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0/0
R1#
R1# ping 172.16.2.2
Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 172.16.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5)
R1# ping 192.168.2.1
Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

IPv6 Starting Routing Tables

- Each router has entries only for directly connected networks and associated local addresses.
- R1 can ping R2, but cannot ping the R3 LAN.

```
R1# show ipv6 route | begin C
C 2001:DB8:ACAD:2::/64 [0/0]
    via Serial0/1/0, directly connected
L 2001:DB8:ACAD:2::1/128 [0/0]
    via Serial0/1/0, receive
C 2001:DB8:ACAD:3::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
L FF00::/8 [0/0]
    via Null0, receive

R1#
R1# ping 2001:db8:acad:2::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2::2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
R1# ping 2001:DB8:cafe:2::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:CAFE:2::1, timeout is 2 seconds:
% No valid route for destination
Success rate is 0 percent (0/1)
```